Subject: Date:

Practice problem

Original Data

age	height	weight	fit / unfit
25	170	68	1
30	165	75	0
22	180	72	1
40	160	85	0
28	175	70	1
35	168	78	0
27	182	74	1
38	158	90	0
29	172	71	1
33	166	80	0

Normalize data for each feature -

$$\text{min_max scalling} = \frac{x_i - x_{min}}{x_{max} - x_{min}}$$

Fatures.

Age		height		weight	
Age_min	Age_Max	h_min	h_max	w_min	w_Max
22	40	158	182	68	90

Formula

$$\frac{x_i(age) - 22}{40 - 22} \Rightarrow \frac{x_i(age) - 22}{18}$$

$$\frac{x_i(he) - 158}{182 - 158} \Rightarrow \frac{x_i(h) - 158}{24}$$

$$= \frac{x_i(w) - 68}{90 - 68} = \frac{x_i(w) - 68}{22}$$

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Scalled Data

Age	height	weight	Pit/unfit	Cos_Dis C)	Pick to $K=\sqrt{n}=$
0.167	0.500	0	1	0.7943	X
0.444	0.292	0.3181	0	0.3238	Pick
0.000	0.0167	0.1818	1	1.1962	X
1.000	0.0833	0.7727	0	1.2281	X
0.333	0.708	0.0909	1	0.7454	X
0.722	0.4167	0.4545	0	0.2985	Pick
0.278	1	0.2727	1	0.9106	X
0.889	0	1	0	1.4277	X
0.389	0.5833	0.1364	1	0.5192	X
0.611	0.333	0.5454	0	0.3621	Pick
x	y	z		D_i	

axis

now calculate the distance for the input:-

Age = 32 ; Height = 169 ; Weight = 76

= 0.556 = 0.4583 = 0.3636

Calculat Distance, Manhattan

$$D_i = |x_i - Age| + |y_i - height| + |z - weight|$$

now, $K = \sqrt{n}$

$= \sqrt{10}$

$= 3$

$n = 10$

So pick first 3 shotest distance

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Bit status of Picked instance

0	0	0

count (0) = 3 | count (1) = 0

Since count (0) > count (1)

$\downarrow$ 3 $\downarrow$ 0

So, Age = 32

height = 169

Weight = 76

Prediction,

bit = [0]